Basic and Diluted EPS: Five Basic Problems

Self-learning workbook with step-by-step solutions

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HOW TO USE THIS WORKBOOK

- Each problem has **one year** of data and only **two share adjustments**. You calculate two things: **basic EPS** and **diluted EPS**.
- Read the problem and try it first. Then follow the solution step by step.
- Every step shows a **Clue** (what to notice and why), the **Formula**, the **Working** with numbers and, where students often slip, a **Watch out** note.
- Amounts are in ₹ crore and shares in crore, so EPS comes out in ₹ per share. The year runs from 1 April 2025 to 31 March 2026 (FY2026).
- Keep full figures while working; round only the final EPS to two decimals.

The ideas you need, in plain words

Share event	What happens	How to count the shares
New issue for cash	New shares are sold at market price; cash comes in.	Count the new shares only from the date of issue (time-weight by months).
Bonus issue	Free shares are given to existing shareholders; no cash comes in.	**Not** time-weighted. Treat the bonus shares as if issued at the start of the year.
Rights issue	Shares are offered to existing shareholders below market price.	Part sale for cash, part free gift. Use the **rights factor** for the months before the issue; count the new shares from the issue date.
Buyback	The company buys back its own shares and cancels them.	Remove the bought-back shares only from the buyback date (time-weight by months).
Conversion	Debentures or preference shares turn into equity shares.	Shares already converted: count from the conversion date (basic EPS). Shares that could still be created: add in diluted EPS.

KEY FORMULAS

Basic EPS = (Profit after tax − Preference dividend) ÷ Weighted average equity shares

Weighted average shares = Σ (Shares outstanding × Months outstanding ÷ 12)

Diluted EPS = (Basic numerator + Add-back) ÷ (Weighted average shares + Potential shares)

Add-back: for convertible debentures = Interest × (1 − Tax rate); for convertible preference shares = Preference dividend (no tax adjustment)

Dilution test: Incremental EPS = Add-back ÷ Potential shares. Include the instrument only if this is **below** basic EPS.

WHY ADD BACK INTEREST AFTER TAX, BUT DIVIDEND IN FULL?

Diluted EPS asks: "What if the convertible had become shares?" Then the company would stop paying interest (or preference dividend).

- **Interest** is an expense that reduces tax. If it stops, profit rises by the interest **minus** the extra tax: Interest × (1 − Tax rate).
- **Preference dividend** is paid out of profit after tax. If it stops, the equity holders keep the whole amount, so add it back in full.

Problem 1: New issue for cash + convertible debentures

THE PROBLEM

Ganga Foods Ltd had 10 crore equity shares on 1 April 2025. On 1 October 2025, it issued 2 crore new equity shares for cash at the market price. Profit after tax for FY2026 (year ended 31 March 2026) was ₹132 crore. The company also has ₹100 crore of 10% convertible debentures outstanding for the whole year, convertible into 2 crore equity shares. The tax rate is 25%. Calculate basic EPS and diluted EPS for FY2026.

WHAT YOU MUST FIND

(a) Basic EPS for FY2026.
(b) Diluted EPS for FY2026.

Try the problem yourself first. Then check your working against the solution, one step at a time.

BEFORE YOU START: THE IDEA AND THE RULE

- **Adjustment 1, new issue for cash:** the 2 crore new shares count only for the months after they were issued.
- **Adjustment 2, convertible debentures:** they do not change basic EPS. For diluted EPS, assume they were converted at the start of the year: add the shares and add back the after-tax interest.

Solution

Step 0. Write down the data given

Item	Value
Shares on 1 April 2025	10 crore
New shares issued for cash on 1 October 2025	2 crore (outstanding 6 months)
Profit after tax	₹132 crore
10% convertible debentures (whole year)	₹100 crore, convert into 2 crore shares
Tax rate	25%

Part (a): Basic EPS

Step 1. Weighted average shares → **11.00 crore**

Clue: The new shares were issued on 1 October, so they were outstanding for 6 months (October to March). The original 10 crore shares were outstanding for all 12 months.

Formula: Weighted shares = Opening shares + New shares × Months outstanding ÷ 12

Working: 10 + 2 × 6/12 = 10 + 1 = 11 crore

Watch out: Do not add the full 2 crore. Check another way: 10 × 6/12 + 12 × 6/12 = 5 + 6 = 11.

Step 2. Basic EPS → **₹12.00**

Clue: There are no preference shares, so the whole profit belongs to equity shareholders.

Formula: Basic EPS = Profit after tax ÷ Weighted average shares

Working: 132 ÷ 11 = ₹12.00

Part (b): Diluted EPS

Step 3. After-tax interest on the debentures → **₹7.50 crore**

Clue: If the debentures convert, the company saves the interest. But interest reduces tax, so profit rises by the after-tax interest.

Formula: After-tax interest = Face value × Interest rate × (1 − Tax rate)

Working: 100 × 10% × (1 − 0.25) = 10 × 0.75 = ₹7.50 crore

Step 4. Dilution test → **Dilutive**

Clue: Include debentures only if they reduce EPS.

Formula: Incremental EPS = After-tax interest ÷ Potential shares; dilutive if below basic EPS

Working: 7.50 ÷ 2 = ₹3.75, which is below ₹12.00, so the debentures are dilutive

Step 5. Diluted EPS → **₹10.73**

Formula: Diluted EPS = (Profit after tax + After-tax interest) ÷ (Weighted average shares + Potential shares)

Working: (132 + 7.50) ÷ (11 + 2) = 139.50 ÷ 13 = ₹10.73

Watch out: The debentures were outstanding for the whole year, so add all 2 crore potential shares (no time-weighting).

FINAL ANSWER

Basic EPS = **₹12.00** Diluted EPS = **₹10.73**

Problem 2: Bonus issue + convertible preference shares

THE PROBLEM

Yamuna Textiles Ltd had 20 crore equity shares on 1 April 2025. On 1 January 2026 it issued bonus shares in the ratio 1:4 (one bonus share for every four shares held). Profit after tax for FY2026 was ₹230 crore. The company has ₹100 crore of 10% convertible preference shares, outstanding for the whole year; the preference dividend of ₹10 crore for the year has been paid. These preference shares are convertible into 4 crore equity shares (a number already adjusted for the bonus issue). Calculate basic EPS and diluted EPS for FY2026.

WHAT YOU MUST FIND

(a) Basic EPS for FY2026.
(b) Diluted EPS for FY2026.

Try the problem yourself first. Then check your working against the solution, one step at a time.

BEFORE YOU START: THE IDEA AND THE RULE

- **Adjustment 1, bonus issue:** no cash comes in, so the bonus shares are **not** time-weighted. They are treated as if issued on 1 April 2025, the start of the year.
- **Adjustment 2, convertible preference shares:** the preference dividend is deducted for basic EPS. For diluted EPS, assume conversion: add the shares and add back the dividend in full (no tax adjustment).

Solution

Step 0. Write down the data given

Item	Value
Shares on 1 April 2025	20 crore
Bonus issue on 1 January 2026	1 bonus share for every 4 held
Profit after tax	₹230 crore
Preference dividend paid	₹10 crore
Convertible preference shares (whole year)	convert into 4 crore shares

Part (a): Basic EPS

Step 1. Bonus shares issued → **5 crore**

Formula: Bonus shares = Shares held × Bonus ratio
Working: 20 × 1/4 = 5 crore

Step 2. Weighted average shares → **25.00 crore**

Clue: A bonus issue gives free shares and brings in no cash, so the business earns the same profit with more shares from the start. The bonus shares are therefore counted for the **full year**, even though they were issued on 1 January.
Formula: Weighted shares = Opening shares + Bonus shares (full year)
Working: 20 + 5 = 25 crore
Watch out: Do **not** use 20 + 5 × 3/12. That would treat the bonus like a new issue for cash.

Step 3. Basic numerator → **₹220 crore**

Clue: The preference dividend belongs to the preference shareholders, so remove it to get the profit for equity shareholders.

Formula: Basic numerator = Profit after tax − Preference dividend

Working: 230 − 10 = ₹220 crore

Step 4. Basic EPS → **₹8.80**

Formula: Basic EPS = Basic numerator ÷ Weighted average shares

Working: 220 ÷ 25 = ₹8.80

Part (b): Diluted EPS

Step 5. Dilution test → **Dilutive**

Clue: If the preference shares convert, no preference dividend is paid, so ₹10 crore comes back to equity holders. Dividends are paid from post-tax profit, so there is no tax adjustment.

Formula: Incremental EPS = Preference dividend ÷ Potential shares; dilutive if below basic EPS

Working: 10 ÷ 4 = ₹2.50, which is below ₹8.80, so dilutive

Step 6. Diluted EPS → **₹7.93**

Formula: Diluted EPS = (Basic numerator + Preference dividend) ÷ (Weighted average shares + Potential shares)

Working: (220 + 10) ÷ (25 + 4) = 230 ÷ 29 = ₹7.93

Watch out: The 4 crore conversion shares are already adjusted for the bonus issue. Do not multiply them again.

FINAL ANSWER

Basic EPS = **₹8.80** Diluted EPS = **₹7.93**

Problem 3: Rights issue + convertible debentures

THE PROBLEM

Krishna Cables Ltd had 12 crore equity shares on 1 April 2025. On 1 October 2025 it made a rights issue of one share for every four shares held, at ₹100 per share. The market price of the share just before the rights issue was ₹125. Profit after tax for FY2026 was ₹165 crore. The company also has ₹100 crore of 10% convertible debentures, outstanding for the whole year and convertible into 1.25 crore equity shares. The tax rate is 25%. Calculate basic EPS and diluted EPS for FY2026.

WHAT YOU MUST FIND

(a) Basic EPS for FY2026.

(b) Diluted EPS for FY2026.

Try the problem yourself first. Then check your working against the solution, one step at a time.

BEFORE YOU START: THE IDEA AND THE RULE

- **Adjustment 1, rights issue below market price:** it is partly a sale for cash and partly a free gift (a bonus element). The gift part is handled with a **rights factor** applied to the months **before** the issue. The new shares count from the issue date.
- **Adjustment 2, convertible debentures:** same as Problem 1. Add the shares and the after-tax interest for diluted EPS.

Solution

Step 0. Write down the data given

Item	Value
Shares on 1 April 2025	12 crore
Rights issue on 1 October 2025	1 new share for every 4 held, at ₹100
Market price before the rights issue (cum-rights price)	₹125
Profit after tax	₹165 crore
10% convertible debentures (whole year)	₹100 crore, convert into 1.25 crore shares
Tax rate	25%

Part (a): Basic EPS

Step 1. Theoretical ex-rights price (TERP) → **₹120.00**

Clue: TERP is the share price we expect after the cheap rights shares are added. "One for every four" means a holder of 4 old shares receives 1 new share.

Formula: TERP = (Old shares × Market price + New shares × Rights price) ÷ (Old shares + New shares)

Working: (4 × 125 + 1 × 100) ÷ (4 + 1) = 600 ÷ 5 = ₹120

Step 2. Rights factor → **1.041667**

Clue: The rights factor measures the free-gift part of the rights issue.

Formula: Rights factor = Market price before the issue ÷ TERP

Working: 125 ÷ 120 = 1.041667

Watch out: Keep the full figure 1.041667. Rounding to 1.04 gives a slightly wrong answer.

Step 3. Rights shares issued → **3 crore**

Formula: Rights shares = Shares held × Rights ratio

Working: 12 × 1/4 = 3 crore, so shares after the issue = 12 + 3 = 15 crore

Step 4. Weighted average shares → **13.75 crore**

Clue: Split the year at 1 October. April to September: 12 crore shares, multiplied by the rights factor. October to March: 15 crore shares.

Formula: Weighted shares = Shares before × Rights factor × Months ÷ 12 + Shares after × Months ÷ 12

Working: 12 × 1.041667 × 6/12 = 6.25

15 × 6/12 = 7.50

Total = 6.25 + 7.50 = 13.75 crore

Watch out: Apply the rights factor only to the months **before** the rights issue.

Step 5. Basic EPS → **₹12.00**

Formula: Basic EPS = Profit after tax ÷ Weighted average shares

Working: 165 ÷ 13.75 = ₹12.00

Part (b): Diluted EPS

Step 6. After-tax interest on the debentures → **₹7.50 crore**

Formula: After-tax interest = Face value × Interest rate × (1 − Tax rate)

Working: 100 × 10% × 0.75 = ₹7.50 crore

Step 7. Dilution test → **Dilutive**

Formula: Incremental EPS = After-tax interest ÷ Potential shares; dilutive if below basic EPS

Working: 7.50 ÷ 1.25 = ₹6.00, which is below ₹12.00, so dilutive

Step 8. Diluted EPS → **₹11.50**

Formula: Diluted EPS = (Profit after tax + After-tax interest) ÷ (Weighted average shares + Potential shares)

Working: (165 + 7.50) ÷ (13.75 + 1.25) = 172.50 ÷ 15 = ₹11.50

FINAL ANSWER

Basic EPS = **₹12.00** Diluted EPS = **₹11.50**

Problem 4: Buyback + convertible preference shares

THE PROBLEM

Narmada Cement Ltd had 50 crore equity shares on 1 April 2025. On 1 October 2025 it bought back 5 crore of its own shares and cancelled them. Profit after tax for FY2026 was ₹390 crore. The company has ₹100 crore of 10% convertible preference shares, outstanding for the whole year; the preference dividend of ₹10 crore for the year has been paid. These preference shares are convertible into 5 crore equity shares. Calculate basic EPS and diluted EPS for FY2026.

WHAT YOU MUST FIND

(a) Basic EPS for FY2026.
(b) Diluted EPS for FY2026.

Try the problem yourself first. Then check your working against the solution, one step at a time.

BEFORE YOU START: THE IDEA AND THE RULE

- **Adjustment 1, buyback:** the cancelled shares stop counting from the buyback date. Before that date they were still outstanding.
- **Adjustment 2, convertible preference shares:** same as Problem 2. Deduct the dividend for basic EPS; add it back in full for diluted EPS.

Solution

Step 0. Write down the data given

Item	Value
Shares on 1 April 2025	50 crore
Shares bought back and cancelled on 1 October 2025	5 crore
Profit after tax	₹390 crore
Preference dividend paid	₹10 crore
Convertible preference shares (whole year)	convert into 5 crore shares

Part (a): Basic EPS

Step 1. Weighted average shares **→ 47.50 crore**

Clue: The 5 crore bought-back shares were outstanding from April to September and cancelled on 1 October, so remove them for the last 6 months only.

Formula: Weighted shares = Opening shares − Shares bought back × Months after buyback ÷ 12

Working: $50 - 5 \times 6/12 = 50 - 2.5 = 47.50$ crore

Watch out: Check another way: $50 \times 6/12 + 45 \times 6/12 = 25 + 22.5 = 47.5$.

Step 2. Basic numerator **→ ₹380 crore**

Formula: Basic numerator = Profit after tax − Preference dividend

Working: 390 − 10 = ₹380 crore

Step 3. Basic EPS → **₹8.00**

Formula: Basic EPS = Basic numerator ÷ Weighted average shares

Working: 380 ÷ 47.5 = ₹8.00

Part (b): Diluted EPS

Step 4. Dilution test → **Dilutive**

Clue: If the preference shares convert, the ₹10 crore dividend is not paid, so it is added back in full.

Formula: Incremental EPS = Preference dividend ÷ Potential shares; dilutive if below basic EPS

Working: 10 ÷ 5 = ₹2.00, which is below ₹8.00, so dilutive

Step 5. Diluted EPS → **₹7.43**

Formula: Diluted EPS = (Basic numerator + Preference dividend) ÷ (Weighted average shares + Potential shares)

Working: (380 + 10) ÷ (47.5 + 5) = 390 ÷ 52.5 = ₹7.43

FINAL ANSWER

Basic EPS = **₹8.00** Diluted EPS = **₹7.43**

Problem 5: Conversion during the year + unconverted debentures

THE PROBLEM

Kaveri Motors Ltd had 40 crore equity shares on 1 April 2025. On that date it also had ₹200 crore of 8% convertible debentures, convertible into 10 crore equity shares in total. On 1 October 2025, half of the debentures (₹100 crore) were converted into 5 crore equity shares. The other half (₹100 crore) remained unconverted at 31 March 2026. Profit after tax for FY2026, after charging the debenture interest actually paid, was ₹255 crore. The tax rate is 25%. Calculate basic EPS and diluted EPS for FY2026.

WHAT YOU MUST FIND

(a) Basic EPS for FY2026.

(b) Diluted EPS for FY2026.

Try the problem yourself first. Then check your working against the solution, one step at a time.

BEFORE YOU START: THE IDEA AND THE RULE

- **Adjustment 1, conversion during the year:** the 5 crore converted shares are real shares from 1 October, so they enter **basic** EPS for 6 months.
- **Adjustment 2, unconverted debentures:** they are potential shares and enter **diluted** EPS for the whole year.
- **The simple idea for diluted EPS:** pretend **all** the debentures were converted on 1 April. Then there would be 40 + 10 = 50 crore shares and no debenture interest at all.

Solution

Step 0. Write down the data given

Item	Value
Shares on 1 April 2025	40 crore
Debentures converted on 1 October 2025	₹100 crore into 5 crore shares
Debentures still unconverted at year end	₹100 crore (would give 5 crore shares)
Interest rate on debentures	8%
Profit after tax (after interest actually paid)	₹255 crore
Tax rate	25%

Part (a): Basic EPS

Step 1. Weighted average shares **→ 42.50 crore**

Clue: The converted shares became real shares on 1 October, so count them for 6 months, just like a new issue.

Formula: Weighted shares = Opening shares + Converted shares × Months outstanding ÷ 12

Working: 40 + 5 × 6/12 = 40 + 2.5 = 42.50 crore

Step 2. Basic EPS → ₹6.00

Clue: The profit given is already after the interest actually paid, so use it directly.

Formula: Basic EPS = Profit after tax ÷ Weighted average shares

Working: 255 ÷ 42.5 = ₹6.00

Part (b): Diluted EPS

Step 3. Potential shares to add → 7.50 crore

Clue: Two pieces. (i) The unconverted half would give 5 crore shares for the whole year. (ii) The converted half is in basic EPS only from 1 October; for diluted EPS we assume it converted on 1 April, so add its 5 crore shares for the 6 months before conversion (April to September).

Formula: Potential shares = Unconverted shares × 12/12 + Converted shares × Months before conversion ÷ 12

Working: 5 × 12/12 + 5 × 6/12 = 5 + 2.5 = 7.50 crore

Watch out: Check: 42.5 + 7.5 = 50 crore = 40 + 10, exactly as if all debentures had converted on 1 April.

Step 4. After-tax interest to add back → ₹9.00 crore

Clue: Add back all the debenture interest that was charged in the year, because under our assumption no interest would have been paid. (i) Unconverted half: interest for the whole year. (ii) Converted half: interest for April to September only (it stopped after conversion).

Formula: After-tax interest = Face value × Interest rate × Months ÷ 12 × (1 − Tax rate)

Working: (i) 100 × 8% × 12/12 × 0.75 = ₹6.00 crore
(ii) 100 × 8% × 6/12 × 0.75 = ₹3.00 crore
Total = 6.00 + 3.00 = ₹9.00 crore

Step 5. Dilution test → Dilutive

Formula: Incremental EPS = After-tax interest ÷ Potential shares; dilutive if below basic EPS

Working: 9.00 ÷ 7.5 = ₹1.20, which is below ₹6.00, so dilutive

Step 6. Diluted EPS → ₹5.28

Formula: Diluted EPS = (Profit after tax + After-tax interest) ÷ (Weighted average shares + Potential shares)

Working: (255 + 9) ÷ (42.5 + 7.5) = 264 ÷ 50 = ₹5.28

FINAL ANSWER

Basic EPS = **₹6.00** Diluted EPS = **₹5.28**

Answers at a glance

No.	Adjustments	Weighted shares (crore)	Basic EPS (₹)	Diluted EPS (₹)
1	New issue for cash + convertible debentures	11.00	12.00	10.73
2	Bonus issue + convertible preference shares	25.00	8.80	7.93
3	Rights issue + convertible debentures	13.75	12.00	11.50
4	Buyback + convertible preference shares	47.50	8.00	7.43
5	Conversion during the year + unconverted debentures	42.50	6.00	5.28

FIVE THINGS TO REMEMBER

- New issue, buyback and conversion: time-weight by the months the shares were outstanding.
- Bonus issue: count the bonus shares for the whole year.
- Rights issue below market price: rights factor for the months before; new shares from the issue date.
- Debentures: add back interest **after tax**. Preference shares: add back the dividend **in full**.
- Include a convertible in diluted EPS only if its incremental EPS is below basic EPS.